Data Visualization Associate Early Remote Internship
Team 06

Deliverable Name: Data Cleaning & Validation Report

Platform: Supabase Cloud PostgreSQL & Looker Studio | Deadline: October 19, 2026

1. Data Integration Process

This section explains how the Opportunity Data (opportunity_master_data, 5,733 rows, 33 columns) and Learner Data (learner_interaction_data, 15,029 rows, 43 columns) datasets were combined.

Join Logic Used

Valid application rows were filtered into applications_clean and joined via an inner join to opportunities in Supabase Cloud PostgreSQL. Specifically, replacing the Learner# prefix in sk with Opportunity# aligns directly with opportunity_id.

SELECT ... FROM public.applications_clean a INNER JOIN public.opportunity_master_data o ON o.opportunity_id::text = 'Opportunity#' || split_part(a.sk, '#', 2);

Key Fields Involved

- **Opportunity Identifier:** opportunity_id (opportunity_master_data): Primary key of the opportunity table with 5,733 unique values.
- **Application Key:** learner_opportunity_sk (sk): Foreign key identifier where the suffix after Learner# maps directly to the opportunity identifier.
- **Learner Identifier:** learner_id (pk): Identifies individual learners, yielding 564 distinct active learners in the integrated view.

Structure of the Integrated Dataset

The integrated dataset (integrated_dataset / integrated_dataset_cleaned) consists of exactly 9,067 records and 25 columns, combining application attributes with opportunity metadata under composite primary key constraints on (learner_id, learner_opportunity_sk).

Dataset Name	Supabase Table	Rows	Columns
Opportunity Data	opportunity_master_data	5,733	33
Learner Data	learner_interaction_data	15,029	43

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2. Data Cleaning Activities

A dedicated working copy (integrated_dataset_cleaned) was created in Supabase to review data quality issues and apply hygiene actions without altering original source tables.

Cleaning Steps Performed

- **Missing-Value Assessment:** Initial profiling revealed missing values in status (17), payment status (34), assigned cohort (3,515), and apply date (21).
- **Duplicate Review:** Duplicate checks using composite learner and opportunity keys confirmed zero duplicate pairs in the integrated table.
- **Status Standardization:** Inconsistent status representations (e.g., 1010 vs 1010.0) were standardized using text replacement to merge fragmented codes.
- **Categorical Normalization:** Profiled location labels (Virtual vs Work From Home) and currency formats (EUR vs EURO).

Assumptions Made

- **Cohort Assumption:** Missing cohort values were retained as NULL because an unassigned cohort does not imply an operational error.
- **Status Suffix Assumption:** Status codes with and without trailing .0 represent identical application stages.

Rationale Behind Major Decisions

Raw NULL values and missing operational dates were preserved as NULL rather than artificially imputed to prevent fabricating metrics. Separate categories like Virtual and Work From Home were preserved because their underlying operational meanings differ.

3. Validation & Comparison

This section compares the dataset before and after cleaning to evaluate quality enhancements.

Improvements Made

Missing status rates improved significantly from 39.78% in raw exports down to 0.19% in the integrated view. Inconsistent .0 status suffixes were completely eliminated.

Changes in Data Quality

The dataset transitioned from unconstrained text columns to robust PostgreSQL data types (timestamp with time zone and numeric).

Data Quality Metric	Original Learner Data	Integrated Dataset	Cleaned Dataset
Total Record Count	15,029	9,067	9,067
Missing Status Count	5,978 (39.78%)	17 (0.19%)	17 (0.19%)
Missing Apply Date	5,982 (39.80%)	21 (0.23%)	21 (0.23%)
Missing Payment Status	Not Assessed	Not Assessed	34 (0.37%)

Validation Checks Performed

- **Integrity Checks:** Confirmed zero unmatched opportunity references and zero negative financial values.
- **Composite Key Validation:** Ran duplicate pair verification queries returning zero duplicate rows.

4. Updated Analysis

This section includes sample SQL queries, summary statistics, and visualizations demonstrating that the dataset is fully ready for analysis and dashboard development.

Summary Metric	Cleaned Integrated Result
Total Cleaned Records	9,067
Unique Learners	564
Unique Opportunities	3,171
Unique Status Values	16
Opportunity Categories	10

Sample SQL Query for Status Distribution:

```
SELECT COALESCE(status, 'Not Specified') AS status, COUNT(*) AS total_records FROM public.integrated_dataset_cleaned GROUP BY status ORDER BY total_records DESC;
```

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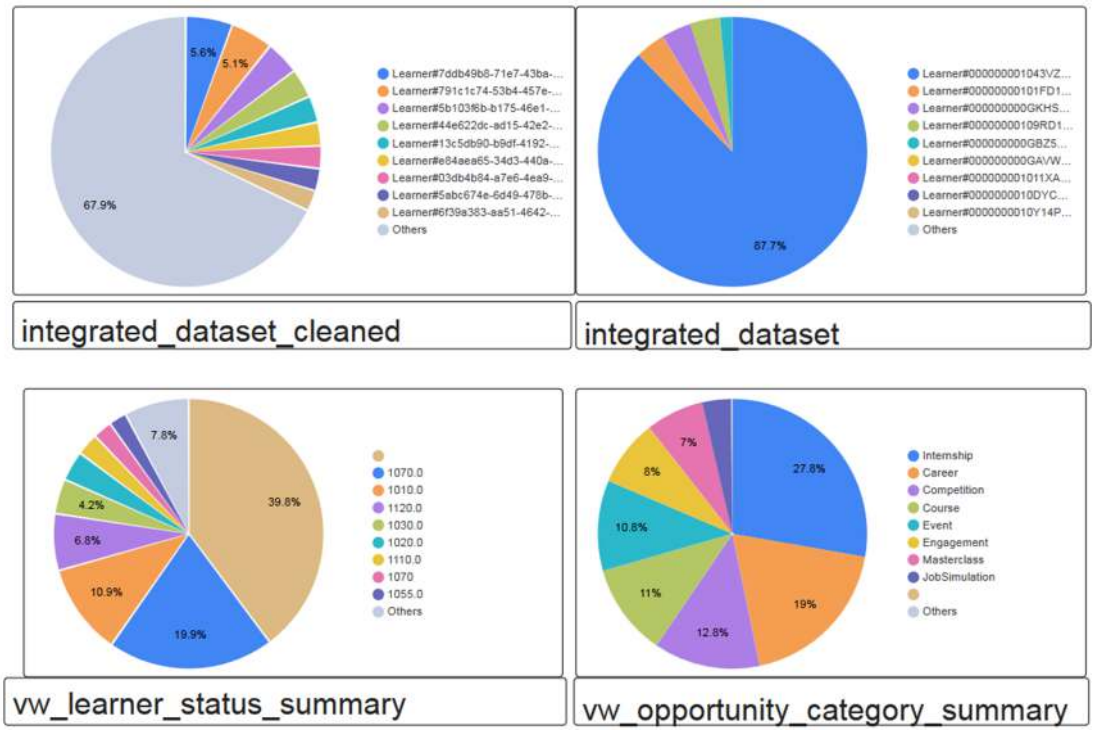


Figure 4.1: Looker Studio Visualizations across integrated_dataset_cleaned, integrated_dataset, vw_learner_status_summary, and vw_opportunity_category_summary.